

§3.3

Multiplication in Other Bases

One table to build, then two methods you already know

Today's Goal

After completing this section, you should be able to:

- Multiply single-digit numbers in a base other than ten by building a multiplication table for that base
- Multiply multi-digit numbers in a base other than ten using the standard algorithm
- Use the Exploding Dots model to multiply numbers in a base other than ten

Learning Target 9

LT.9 Different Number Bases

I can convert numbers from base ten to base n and vice versa for any whole number $n > 1$. I can add, subtract, multiply, and divide in different number bases.

What Base Is This?

Nobody wrote down the base. Work it out, then finish the table.

$\times$	0	1	2	3	4
0					
1					
2				11	
3			11		22
4				22	31

The Base-5 Multiplication Table

$\times$	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	11	13
3	0	3	11	14	22
4	0	4	13	22	31

Row 0 and row 1 cost nothing, and the table is symmetric, so 2×3 and 3×2 are the same work.

That leaves six products: 2×2 , 2×3 , 2×4 , 3×3 , 3×4 , 4×4 .

Every Product on Its Own Row

Evaluate $24_{\text{five}} \times 342_{\text{five}}$ using the standard algorithm.

Each product is written on its own: its ones digit on the product row, and the number of fives it makes just above it, one column to the left.

				3	4	2
				2	4	
×				2	1	
		2	3	2	1	3
	1	1	0			
+		1	3	4		
	2	0	4	1	3	

So $24_{\text{five}} \times 342_{\text{five}} = 20413_{\text{five}}$.

Your Turn

Evaluate $32_{\text{five}} \times 214_{\text{five}}$ using the standard algorithm.

$$\begin{array}{r}
 214 \\
 32 \\
 \hline
 001 \\
 423 \\
 102 \\
 \hline
 + 1132 \\
 \hline
 13003
 \end{array}$$

So $32_{\text{five}} \times 214_{\text{five}} = 13003_{\text{five}}$.

Multiplying on the Machine

Evaluate 152×2 using the Exploding Dots model.

			$1 \leftarrow 10$
152			
$\times 2$			
After multiplying			
After regrouping			

Each dot becomes two, then every ten regroups left. So $152 \times 2 = 304$.

Numbers in Place of Dots

Evaluate 152×12 using the Exploding Dots model.

$1 \leftarrow 10$

152		1	5	2
$\times 12$			1	2
First partial product		2	10	4
Second partial product	1	5	2	0
After adding	1	7	12	4
After regrouping	1	8	2	4

So $152 \times 12 = 1,824$.

Only the Last Step Changes

Evaluate $24_{\text{five}} \times 342_{\text{five}}$ using the Exploding Dots model.

$1 \leftarrow 5$

342_{five}			3	4	2
$\times 24_{\text{five}}$				2	4
First partial product			12	16	8
Second partial product		6	8	4	0
After adding		6	20	20	8
After regrouping	2	0	4	1	3

So $24_{\text{five}} \times 342_{\text{five}} = 20413_{\text{five}}$, as in Example 1.